

Roll No

JC Bose University of Science & Technology YMCA, Faridabad

B. Tech 4th Semester Mechanical Engineering

MANUFACTURING PROCESSES (PCC-ME-405/21)

1st Sessional Test, March 2025

Max. Marks: 15

Time Allowed: 90 minutes

Note: Attempt any three questions.

Q. No.	Statement	Marks	CO
1	Explain the different types of pattern.	5	CO1
2	The total solidification times of three casting shapes are to be compared: (1) a sphere with diameter = 10 cm, (2) a cylinder with diameter and length both = 10 cm, and (3) a cube with each side = 10 cm. The same casting alloy is used in the three cases. (a) Determine the relative solidification times for each geometry. (b) Based on the results of part (a), which geometric element would make the best riser? (c) If the mold constant = 3.5 min/cm^2 in Chvorinov's rule, compute the total solidification time for each casting.	5	CO1
3	Explain the working principle of Rolling with a neat sketch. Also explain the rolling stand arrangement.	5	CO2
4	Explain principle and working parts of Shaper machine by neat sketch.	5	CO3